

SEMINAR III · WEEK 4 · FOUNDATIONS

Fractions & Decimals

Learn → Practice → Check → Analyze → Retry → Apply

Two teaching sessions this week (fractions, then decimals) plus an apply/retry day — not five separate full lessons.

SESSION A

Fractions: Meaning to Operations

What a fraction means · comparing them · adding, multiplying, dividing

WHAT IS A FRACTION?

A fraction is also a division.

numerator

denominator

on a number line

equivalent fractions

Show visuals before procedures — fraction bars and number lines, not just the algorithm.

COMMON MISCONCEPTION

**Is $1/8$ greater than $1/4$,
because $8 > 4$?**

No — more pieces means each piece is smaller.

COMMON MISCONCEPTION

Does simplifying make a fraction smaller?

$$6/8 = 3/4$$

The representation changes. The quantity doesn't.

ADDING FRACTIONS

Same-size pieces first.

Wrong: $1/2 + 1/3 = 2/5$. The pieces must be rewritten with a common denominator first.

Like denominators: add numerators directly. Unlike denominators: find a common denominator first.

MULTIPLYING FRACTIONS

No common denominator needed.

$\frac{2}{3} \times \frac{3}{5} \rightarrow$ multiply numerators, multiply denominators

A common misconception is hunting for a common denominator here — that's an addition rule, not a multiplication one.

DIVIDING FRACTIONS

"Keep, change, flip" — but ask why first.

Before the shortcut, ask: what does this division actually mean? Watch for flipping the wrong fraction or flipping both.

COMMON MISCONCEPTION

Must a fraction be less than 1?

$5/4$

$7/3$

$10/5$

No — improper fractions are legitimate values, not "wrong" answers to fix.

EXIT TICKET — SESSION A

1. Simplify a fraction
2. Compare two fractions
3. Explain why two fractions are equivalent

SESSION B

Decimals & Choosing a Representation

Decimal place value and operations · fractions and decimals as the same number, different form

COMMON MISCONCEPTION

**Is 0.475 greater than 0.8,
because $475 > 8$?**

No — place value decides, not digit count.

0.800 > 0.475. Line up decimal points to compare.

COMMON MISCONCEPTION

Right digits, wrong decimal placement.

2.4×3 — a student answers 72

Estimate first: $2 \times 3 \approx 6$. An answer of 72 should immediately look wrong.

ACTIVITY — SAME NUMBER, DIFFERENT FORM

Match the equivalent cards.

$\frac{1}{2}$

0.5

$\frac{5}{10}$

$\frac{3}{4}$

0.75

$\frac{75}{100}$

WHICH FORM IS MOST USEFUL?

**Fraction, decimal, or percent —
it depends on the context.**

recipes

money

measurement

calculator comparisons

EXIT TICKET — SESSION B

1. Convert one fraction to a decimal
2. Convert one decimal to a fraction
3. Explain which form is more useful in a given context

APPLY, ANALYZE, RETRY

Mixed Practice & Week 4 Check

Independent work, not new instruction.

- Mixed retrieval: fraction meaning, comparison, operations, decimal operations, conversions, reasonableness
- Students classify missed items by error category, then retry after feedback
- Week 4 check, then reflection: one strength, one developing skill, one error pattern, one strategy for next week

TRANSITION TO WEEK 5

Percent

Fraction $\leftrightarrow$ Decimal $\leftrightarrow$ Percent, e.g. $1/2 = 0.5 = 50\%$. Applied to discounts, tips, tax, wages, and grades.